FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

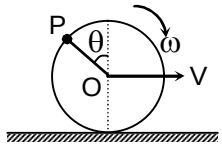
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

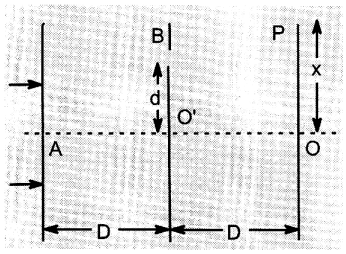
1. A disc of radius *R* rolls on a horizontal surface with linear velocity *V* and angular velocity ω. There is a point *P* on circumference of disc at angle θ with upward vertical diameter measured anticlockwise see figure, which has a vertical velocity. Here θ is equal to



(A) $\pi + \sin^{-1}\dfrac{V}{R\omega}$ (B) $\dfrac{\pi}{2} - \sin^{-1}\dfrac{V}{R\omega}$

(C) $\pi - \cos^{-1}\dfrac{V}{R\omega}$ (D) $\pi + \cos^{-1}\dfrac{V}{R\omega}$

2. The minimum value of d so that there is a dark fringe at O is d_{min}. For the value of d_{min}, the distance at which the next bright fringe is formed is x. Then:

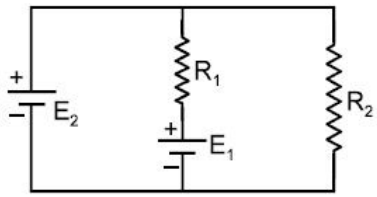


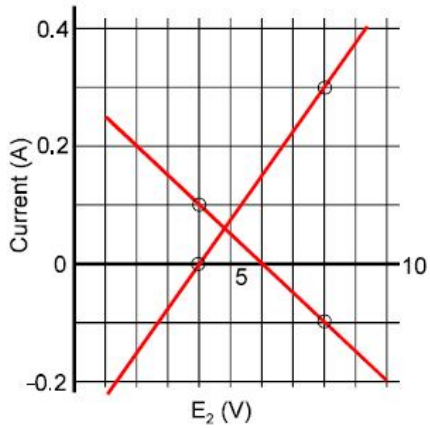
(A) $d_{min} = \sqrt{\lambda D}$ (B) $d_{min} = \sqrt{\dfrac{\lambda D}{2}}$

(C) $x = \dfrac{d_{min}}{2}$ (D) $x = d_{min}$

Space for Rough work

3. In the circuit given below, both batteries are ideal. EMF E_1 of battery 1 has a fixed value, but emf E_2 of battery 2 can be varied between 1.0 V and 10.0 V. The graph gives the currents through the two batteries as a function of E_2, but are not marked as which plot corresponds to which battery. But for both plots current is assumed to be negative when the direction of the current through the battery is opposite the direction of that battery's emf. (direction of emf is from negative to positive)





The value of emf E_1 is

(A) 8 V (B) 6 V

(C) 4 V (D) 2 V

4. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.

(A) It the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.

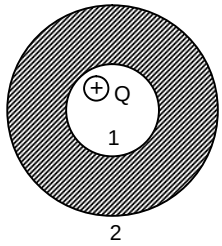
(B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.

(C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.

(D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.

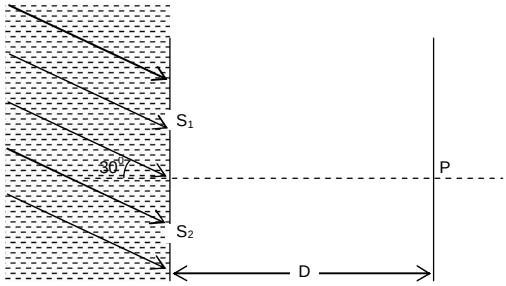
Space for Rough work

5. The positive charge Q is placed in the cavity of the spherical conductor.



(A) the induced charge at the surface 1 is –Q and on the surface 2 is +Q
(B) the surface charge density at inner surface is uniform
(C) the surface charge density at the outer surface is uniform
(D) the charge Q does not experience electric force if it is placed at the centre of the spherical cavity.

6. The given figure shows a YDSE apparatus. Parallel monochromatic coherent light rays are incident on slits S_1 and S_2 ($S_1S_2 = \frac{2}{3}$mm) at an angle 30° with the horizontal. The medium on left side of the slits is water $\left(\mu_w = 4/3\right)$. To obtain the central maxima at point P, a glass slab $\left(\mu_g = 3/2\right)$ inside water is introduced in front of slit S_1. The thickness of the glass slab required for this purpose is



(A) 2mm (B) 4/3 mm
(C) 4 mm (D) 8/3 mm

7. The electron in the hydrogen atom make a transition from an excited state to the ground state. Which of the following statement is true?
(A) Its kinetic energy increases and its potential and total energies decrease
(B) Its kinetic energy decreases, potential energy increases and its total energy remains the same
(C) Its kinetic and total energies decrease and its potential energy increases
(D) Its kinetic potential and total energies decreases

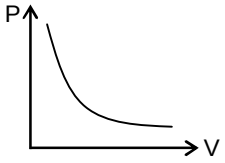
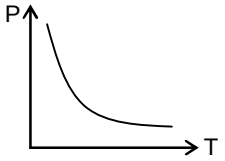
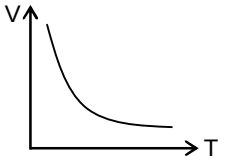
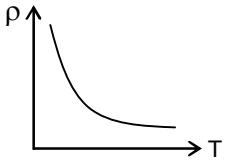
Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

Column–1 shows rectangular hyperbola graphs, Column–2 shows molar heat capacities for monoatomic gases and Column-3 shows work done when process goes from initial state 1 to final state 2.

Column –1	Column –2	Column –3
(I) 	(i) $C = \frac{7R}{2}$	(P) $\Delta W = nR(T_2 - T_1)$
(II) 	(ii) $C = \frac{R}{2}$	(Q) $\Delta W = -nR(T_2 - T_1)$
(III) 	(iii) $C = \frac{5R}{2}$	(R) $\Delta W = 2nR(T_2 - T_1)$
(IV) 	(iv) $C = \infty$	(S) $\Delta W = nRT \ln\frac{V_2}{V_1}$

8. Which option matches the data for an isobaric process?

(A) (IV) (iv) (P) (B) (III) (iii) (P)

(C) (IV) (iii) (P) (D) (III) (i) (Q)

Space for Rough work

9. Which of the following options is a correct combination?
(A) (II) (iii) (R) (B) (III) (ii) (Q)
(C) (III) (iii) (Q) (D) (II) (ii) (Q)

10. Which option matches the data for a process whose $\Delta U = 0$?
(A) (IV) (ii) (S) (B) (II) (iv) (S)
(C) (I) (ii) (Q) (D) (I) (iv) (S)

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

If m gm of ice at θ_1°C is mixed with 100 gm of water at 20°C, the mixture contains m_1 gm of ice and m_2 gm water at θ_2 °C in equilibrium. (Specific heat of water = 1 cal/gm/°C, Specific heat of ice = 0.5 cal/gm/°C, L_{ice} = 80 cal/ gm.

Column -1	Column -2	Column -3
(I) m = 25 gm & θ_1 = 0°C	(i) m_1 = 0	(P) θ_2 = 0°C
(II) m = 20 gm & θ_1 = –10°C	(ii) m_1 = 38.75 gm	(Q) θ_2 = 2.5°C
(III) m = 60 gm & θ_1 = –10°C	(iii) m_1 = 100 gm	(R) m_2 = 120 gm
(IV) m = 100 gm & θ_1 = –40°C	(iv) m_1 = 30 gm	(S) m_2 = 125 gm

11. Which of the following options is a correct combination?
(A) (II) (i) (S) (B) (I) (i) (R)
(C) (II) (i) (Q) (D) (I) (iv) (Q)

12. Which of the following options is a correct combination?
(A) (III) (iv) (P) (B) (III) (ii) (P)
(C) (II) (iii) (P) (D) (II) (ii) (P)

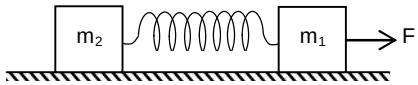
13. Which of the following options is a correct combination?
(A) (IV) (iii) (P) (B) (IV) (iv) (P)
(C) (III) (iv) (Q) (D) (I) (ii) (P)

Space for Rough work

SECTION – C
(Single digit integer type)

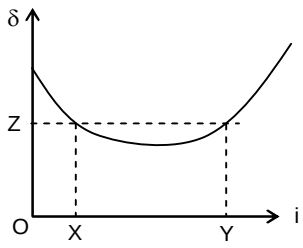
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. Two blocks of masses m_1 = 1kg and m_2 = 2kg are connected by a non deformed light spring. They are lying on a rough horizontal surface. The coefficient of friction between the blocks and the surface is 0.4, what minimum constant force F (In N) has to be applied in horizontal direction to the block of mass m_1 in order to shift the other block? (g = 10 m/s^2)



15. Potential energy of a particle moving along x-axis is given by U = $\frac{x^3}{3}-\frac{9x^2}{2}+20x$. Find out position of stable equilibrium state.

16. Graph shown is between deviation (δ) and angle of incidence I then angle of prism is (in degree) [x = 30°, y = 32° & z = 60°]



17. The period of oscillation of a simple pendulum is given by T = $2\pi\sqrt{\frac{\ell}{g}}$ where I is about 100 cm and is known to have 1mm accuracy. The period is about 2s. The time of 100 oscillations is measured by a stop watch of least count 0.1 s. The percentage error in g is x% then 10x is

18. In an experiment the angle are required to be measured using an instrument. 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half – a – degree (= 0.5^o) then the least count of the instrument is (in minutes)

Space for Rough work

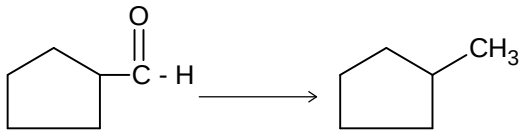
Chemistry

PART – II

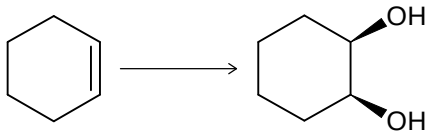
SECTION – A
(One or More than one correct type)

This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following concentrated terms are temperature independent?
(A) Molarity (B) Molality
(C) Mole fraction (D) PPM

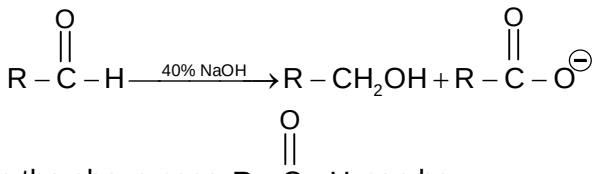
20. 

The above transformation can be carried out by following methods?
(A) Zn(Hg)/Conc HCl (B) NH_2NH_2, OH^-
(C) CH_2SH / CH_2SH, Dry HCl and Raney Ni, Δ (D) $LiAlH_4$

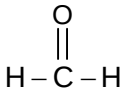
21. 

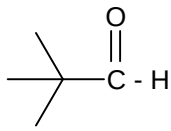
The above transformation can be carried out by the following methods.
(A) Bayer's reagent (B) $OsO_4/NaHSO_3$
(C) CH_3COOOH, H_3O^+ (D) m-CPBA, OH^-

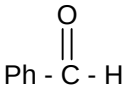
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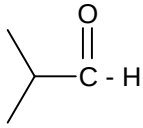
22. 

In the above case can be

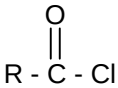
(A) 

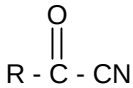
(B) 

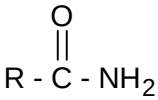
(C) 

(D) 

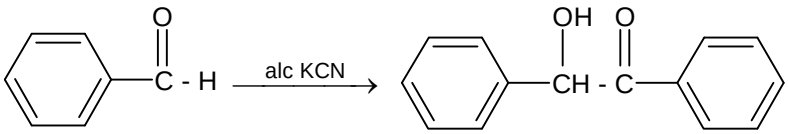
23. Which of the following compounds are more reactive than  towards attack of nucleophile?

(A) 

(B) 

(C) 

(D) 

24. 

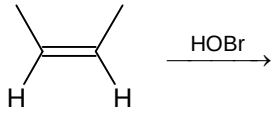
In the above reaction alcohol KCN performs the following function?

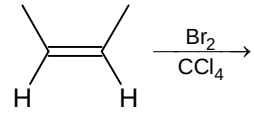
(A) Nucleophile (B) Base

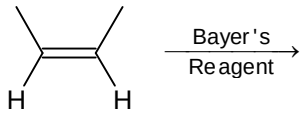
(C) Electron withdrawing (D) Leaving group

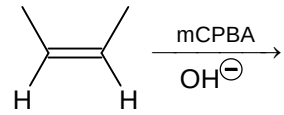
Space for Rough work

25. Which of the following reactions are stereospecific as well as stereoselective?

(A) 

(B) 

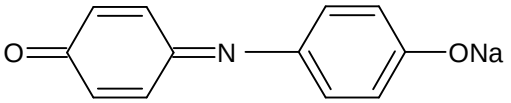
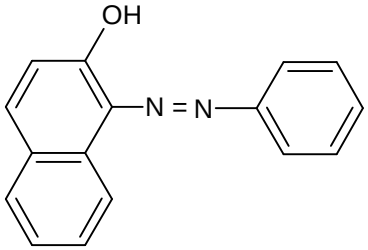
(C) 

(D) 

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

Column – 1 Complexes		**Column – 2 Colour**		**Column – 3 Test**	
(I)	$Na_4[Fe(CN)_5NOS]$	(i)	Red colour	(P)	Liberman's nitroso test
(II)	$[Fe(H_2O)_5NO]SO_4$	(ii)	Blue colour	(Q)	Dye test
(III)		(iii)	Brown colour	(R)	Lassaigne's test
(IV)		(iv)	Violet colour	(S)	Ring test

26. Which of the following pair is correct for detection of phenolic group?

(A) (I), (ii) & (R) (B) (III), (i) & (P)

(C) (IV), (i) & (Q) (D) (II), (iii) & (S)

Space for Rough work

27. Which of the following pair is correct for detection of aromatic amine?
(A) (IV), (i) & (Q) (B) (III), (ii) & (P)
(C) (II), (iii) & (R) (D) None of these

28. Which of the following is used for NO_3^- group detection?
(A) (II), (iii) & (S) (B) (III), (ii) & (Q)
(C) (IV), (iv) & (P) (D) None of these

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column – 1 Order		**Column – 2 $t_{½}$**		**Column – 3 Rate constant**	
(I)	Zero	(i)	$\alpha[\ ]$	(P)	$\frac{2.303}{t}\log\frac{r_0 - r_\infty}{r_t - r_\infty}$
(II)	1st	(ii)	$\alpha[\ ]^{-1}$	(Q)	$\frac{2.303}{t}\log\frac{V_\infty - V_0}{V_\infty - V_t}$
(III)	2nd	(iii)	$\alpha[\ ]^{-2}$	(R)	$\frac{2.303}{t}\log\frac{V_0}{V_t}$
(IV)	3rd	(iv)	$\alpha[\ ]^{0}$	(S)	None of the above

29. Which of the following pair is correct for saponification?
(A) (I), (i) & (S) (B) (II), (iv) & (S)
(C) (III), (ii) & (S) (D) (IV), (iii) & (S)

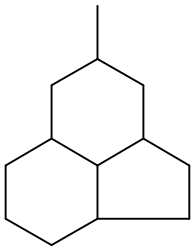
30. Which of the following is correct for ester hydrolysis in acidic medium?
(A) (I), (i) & (Q) (B) (II), (iv) & (Q)
(C) (II), (iv) & (R) (D) (III), (ii) & (S)

31. Thermal decomposition of H_2O_2 has correct combination as follows.
(A) (II), (iv) & (Q) (B) (I), (iv) & (R)
(C) (III), (ii) & (S) (D) None of these

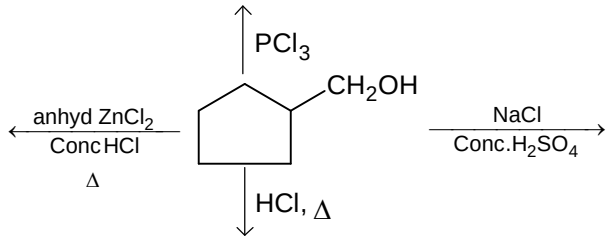
Space for Rough work

SECTION – C
(Single digit integer type)

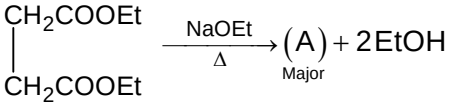
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. 

Total how many chiral C atoms are present in this compound?

33. 

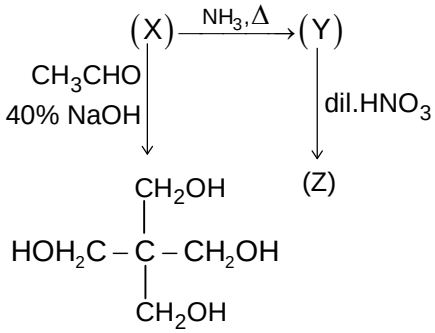
In how many above reaction ring expansion takes place.

34. 

Total how many mole of Grignard's reagent are required to completely react with (A).

Space for Rough work

35.



Total how many sp^2 hybridised N are present in (Z)

36. Determine 2x – y – z where x is the no. of mole of HIO_4 required to oxidize the 1 mole of Fructose, y is the no. of mole of HCOOH produced and z is the no. of mole of HCHO produced.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **7** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Consider the functions $f(x)$ and $g(x)$, both defined from $R \to R$ and are defined as $f(x) = 2x - x^2$ and $g(x) = x^n$ where $n \in N$. If the area between $f(x)$ and $g(x)$ in first quadrant is $\frac{1}{2}$ then n is a divisor of

(A) 12 (B) 15
(C) 20 (D) 30

38. A function $f : R \to R$ is given by $f(x) = \begin{cases} x^4\left(2 + \sin\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$, then

(A) f has a continuous derivative $\forall\, x \in R$ (B) f is a bounded function
(C) f has an global minimum at x = 0 (D) f" is continuous $\forall\, x \in R$

39. Identify correct statement(s) about conic $\sqrt{(x-5)^2 + (y-7)^2} + \sqrt{(x+1)^2 + (y+1)^2} = 12$.

(A) centre is (2, 3) (B) hyperbola with foci (5, 7) and (–1, –1)
(C) ellipse with major axis $4x - 3y + 1 = 0$ (D) eccentricity is $\frac{5}{7}$

40. If $b^2 \geq 4ac$ for the equation $ax^4 + bx^2 + c = 0$, then all roots of the equation will be real, if:

(A) $b > 0, a > 0, c > 0$ (B) $b < 0, a > 0, c > 0$
(C) $b > 0, a > 0, c < 0$ (D) $b > 0, a < 0, c < 0$

41. If a, b, c, d are distinct positive numbers in A.P., then

(A) $ad < bc$ (B) $a + c < b + d$
(C) $a + d = b + c$ (D) $(a+1)(d+1) < (b+1)(c+1)$

Space for Rough work

42. The last digit of $3^{3^{4n}}+1, n \in N$ is:

(A) ${}^4C_3$ (B) ${}^8C_7$

(C) 8 (D) 4

43. In a class tournament when the participants were to play one game with another, two class players fell ill, having played 3 games each. If the total number of games played is 84, the number of participants at the beginning was:

(A) 15 (B) 30

(C) ${}^6C_2$ (D) 48

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Columns –1 contains different functions. Column –2 and Column –3 contains their domain and range respectively.

Column -1	Column -2	Column -3
(I) For the function $f(x)=\ell n\left(x^2-6x+5\right)$	(i) $x \in [-2,-1] \cup [1,2]$	(P) $\left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$
(II) $f(x)=\frac{\{x\}}{1+\{x\}}$	(ii) $x \in (-\infty, 1) \cup (5, \infty)$	(Q) R
(III) $f(x)=\sin\left(2\pi\left(\frac{\sqrt{x+7}-4}{x-9}\right)\right)$	(iii) $x \in R$	(R) $\left[0, \frac{1}{2}\right)$
(IV) $f(x)=\sqrt{4-x^2}+\sqrt{x^2-1}$	(iv) $x \in [-7, \infty)-\{9\}$	(S) $\left[\sqrt{3}, \sqrt{6}\right]$

44. Which of the following is correct combination?

(A) (III) (ii) (P) (B) (IV) (i) (Q)

(C) (II) (iii) (R) (D) (I) (ii) (S)

Space for Rough work

45. Which of the following is incorrect combination?
(A) (I) (ii) (Q) (B) (II) (iii) (R)
(C) (III) (iv) (P) (D) (IV) (i) (P)

46. Which of the following is correct combination?
(A) (I) (iii) (R) (B) (IV) (i) (S)
(C) (II) (iii) (S) (D) (IV) (i) (P)

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

In all these cases $p,q \in N$ and also co – prime numbers.		
Column -1	**Column -2**	**Column -3**
(I) $P(x) = x^4 + ax^3 + bx^2 + cx + d$ where $a,b,c,d \in R$. Suppose $P(0) = 6, P(1) = 7, P(2) = 8$ and $P(3) = 9$ then $P(4) - 4 = \frac{p}{q}$.	(i) p = 19	(P) q = 1
(II) If the sum $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{2}{n^2+n+4}\right) = \tan^{-1}\left(\frac{q}{p}\right)$	(ii) p = 16	(Q) q = 8
(III) Let $P = 1^{-4} + 2^{-4} + 3^{-4} + 4^{-4} + ...\infty$ and $q = 1^{-4} + 3^{-4} + 5^{-4} + 7^{-4} + ...\infty$ then $\frac{p}{q}$	(iii) p = 1	(R) q = 2
(IV) Let $S_1, S_2, S_3, ...$ and $t_1, t_2, t_3 .....$ are two arithmetic sequence such that $S_1 = t_1 \neq 0$, $S_2 = 2t_2$ and $\sum_{i=1}^{10} S_i = \sum_{i=1}^{15} t_i$ then $\frac{S_2 - S_1}{t_2 - t_1} = \frac{p}{q}$	(iv) p = 30	(S) q = 15

47. Which of the following is correct combination?
(A) (IV) (i) (R) (B) (III) (ii) (S)
(C) (I) (iii) (P) (D) (II) (iii) (P)

Space for Rough work

48. Which of the following is incorrect combination?
(A) (III) (ii) (S) (B) (II) (iii) (R)
(C) (IV) (i) (Q) (D) (I) (ii) (P)

49. Which of the following is correct combination?
(A) (I) (iv) (P) (B) (II) (iii) (S)
(C) (IV) (i) (R) (D) (II) (i) (P)

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If $\int_0^1 \frac{dx}{(1+x)(2+x)\sqrt{x(1-x)}} = \frac{\pi A}{\sqrt{6}\left(\sqrt{3}+1\right)}$ then A is equal to

51. Let $f(x) = \frac{e^x - e^{-x}}{2}$ and if $f(g(x)) = x$ then $\frac{1}{501} g\left(\frac{e^{1002}-1}{2e^{501}}\right)$ is equal to

52. If $y = \left(1+\frac{1}{x}\right)^x$ then $\frac{2\sqrt{y_2(2)+\frac{1}{8}}}{\left(\log\frac{3}{2}-\frac{1}{3}\right)}$ is equal to

53. A coin is tossed n times. The probability of getting at least one head is greater than that of getting at least two tails by $\frac{5}{32}$. Then n is:

54. If l_1 and l_2 be the length of the focal chord other than latus rectum of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ which are perpendicular to each other, then $12\left(\frac{l}{l_1}+\frac{l}{l_2}\right) =$

Space for Rough work